Kishan Madlani

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About

Computer Vision Research Engineer specializing in generative and multimodal AI, with hands-on experience building diffusion models from scratch, text-conditioned image generation, and synthetic-data pipelines. Research experience includes latent diffusion, multimodal learning, metric learning, and LLM/RAG systems, with interests in generative modeling, VLMs, and efficient AI.

Experience

Computer Vision Research Engineer Feb 2026 – Aug 2026
CVIT Lab, IIIT Hyderabad *Hyderabad, Telangana*

- Built a diffusion-based model that generates realistic Hindi handwriting mimicking real writers' styles, and used it to create a fully synthetic dataset; combining synthetic and real data improved OCR accuracy by **17%** over real-only training [Tech Report]

Projects

Diffusion-Based Handwriting Generation | *Diffusion Models, Proxy-NCA Metric Learning, PyTorch*

- Built and trained a Stable Diffusion-style latent diffusion model **from scratch** (custom UNet, dual MobileNetV2 style/content encoders, cross-attention Blender) to generate realistic Hindi handwriting in a target writer's style from arbitrary input text, trained on **900 writers and 25K+ samples** with a **270M-parameter** model
- Achieved **FID 17.0** and **HWD 0.97** by designing a style encoder with Proxy-NCA metric learning and column/row masking to disentangle writer identity from text content, decoupling style from content in the diffusion conditioning signal
- GitHub | HF Demo (CPU) | Colab (GPU)

Diffusion-Based Face Image Generation | *PyTorch, Diffusion Models, Latent Space, Image Generation*

- Developed an unconditional face image generation model by training a UNet-based latent diffusion model **from scratch** on the CelebA dataset, achieving a Fréchet Inception Distance (FID) of **12.0**, demonstrating high-fidelity and diverse synthetic face generation
- Model size: **135M parameters**; trained with a standard DDPM noise schedule in a pretrained VAE's compressed latent space to significantly reduce compute vs. pixel-space diffusion
- GitHub | HF Demo

RAG-Based YouTube Video Chatbot | *RAG, FAISS, Pinecone, LangChain, Python, LLMs*

- Built a RAG-based chatbot that enables natural language interaction with YouTube videos using transcripts
- Implemented a vector-based retrieval pipeline using **FAISS** and **Pinecone**, integrating retrieved context with an LLM for grounded response generation
- Deployed chatbot on university Discord server, processing 500+ queries from 47+ users
- Github

Technical Skills

Languages: Python, Java
Deep Learning Frameworks: PyTorch, TensorFlow
Computer Vision: OpenCV, Image Processing, Feature Extraction, CNNs, Vision Transformers (ViTs), Object Detection, Image Generation
Generative AI: GANs, VAEs, Diffusion Models, Latent Diffusion
LLM / Multimodal AI: LLMs, Retrieval-Augmented Generation (RAG), LangChain, LangGraph
Tools: Git, Linux, Weights & Biases (W&B), Optuna

Education

Sardar Vallabhbhai National Institute of Technology (NIT Surat) Surat, India
M.Sc. (Integrated) Physics *Dec 2021 – June 2026*
Thesis: [Image generation: Diffusion-based Models] under Prof. C. V. Jawahar

Relevant Coursework: Machine Learning, Statistics, linear algebra, computer vision, Python